



PennState
College of Information
Sciences and Technology

Lab Assignment Report

COURSE:	IST 894
ASSIGNMENT:	LAB ASSIGNMENT REPORT 12
SUBMITTED BY:	BRANTLEY PRICE
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General Context

This cyber range exercise was entitled “Digital Forensics Techniques.” One of the key learning objectives for the student to grasp was the digital forensics life cycle, which consists of identification, preservation, analysis, documentation, and presentation (Gonzalez, 2024). The student captured the data from an interface, sifted through the data to gather concrete evidence, and hashed the data to ensure its integrity was maintained. This will ultimately allow the information to be easily reconstructed and will tell the story of what happened on the network if the need arises.

Another takeaway, as noted in previous exercises, is the use of legacy protocols like Telnet. Isolating this traffic provides the student a look into how cleartext data can be easily seen using Wireshark via unencrypted connections. The use of Telnet, outside of very niche purposes, is not recommended for any practical reason (University of Pittsburgh, 2025).

As mentioned above, the exercise also highlights the integrity of evidence gathered via forensics techniques. It does so via hashing techniques to create a chain-of-custody and the ability to track evidence tampering. Using hashing techniques creates a traceable record that will stand up to rigorous legal scrutiny if the need arises. Once a file is hashed, any changes to the file will change the file's hash and will clearly show that the file has been tampered with.

Technical Context

Using the command line interface (CLI), the student targeted the *ens32* interface and ran *tcpdump* with elevated privileges, writing packets directly to the *data.pcap* file. The *.pcap* file type is widely used in the network security field as the standard for packet capturing (SolarWinds, n.d.). Additionally, a *timeout* of 20 seconds was used to control the scope and resource use. Using a timeout bounds the packet capture and makes it more reliable for specific timeframes.

Once the packets were captured, the student moved to Wireshark to review the packet contents. Filters, such as *tcp.port == 23* showed traffic that was being passed via the Telnet protocol. Digging deeper, the student searched for *tcp contains login* and found clear text authentication exchanges. This shows the lack of security in legacy protocols like Telnet. Readable prompts and banners between the two endpoints provided evidence as to the who, what, and when of the communication between the two. This is good for someone conducting forensics, but terrible for network security practitioners.

A key portion of this lab centered around the integrity and reproducibility of the data that was captured in the *.pcap* file. Using *md5sum*, *sha256sum*, and *shasum -a <algo>* produced hash digests for *data.pcap*. These hashes were captured before and after the handling and proved that the article did not change. This is a key point: hash values are deterministic and are a necessary means to verify the authenticity of a file (Callaghan, n.d.). If going into a court of law, screenshots will not be enough. Evidence must exist that the file was not changed before or after the capture.

Screenshots

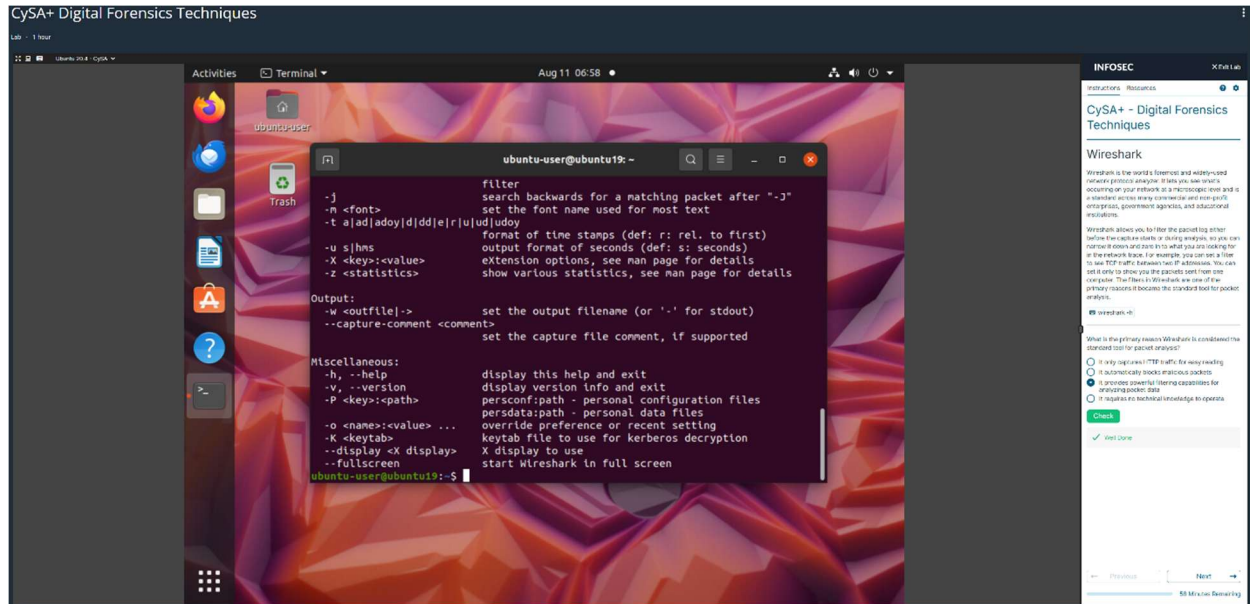


Figure 1: Digital Forensics Techniques and Wireshark Introduction

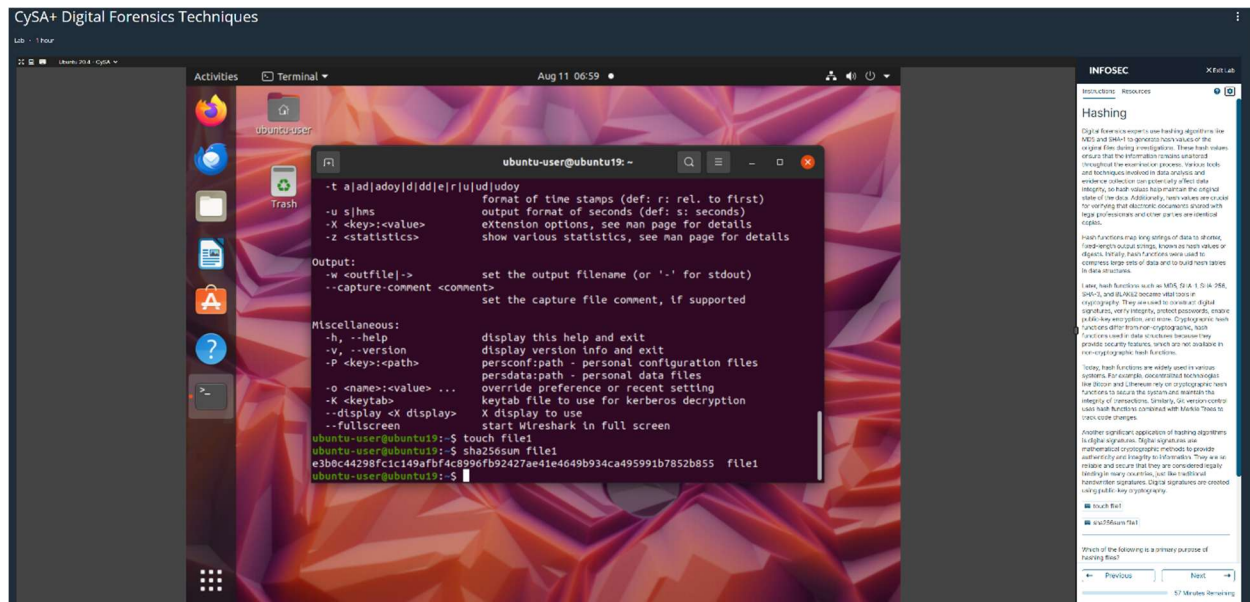


Figure 2: Hashing

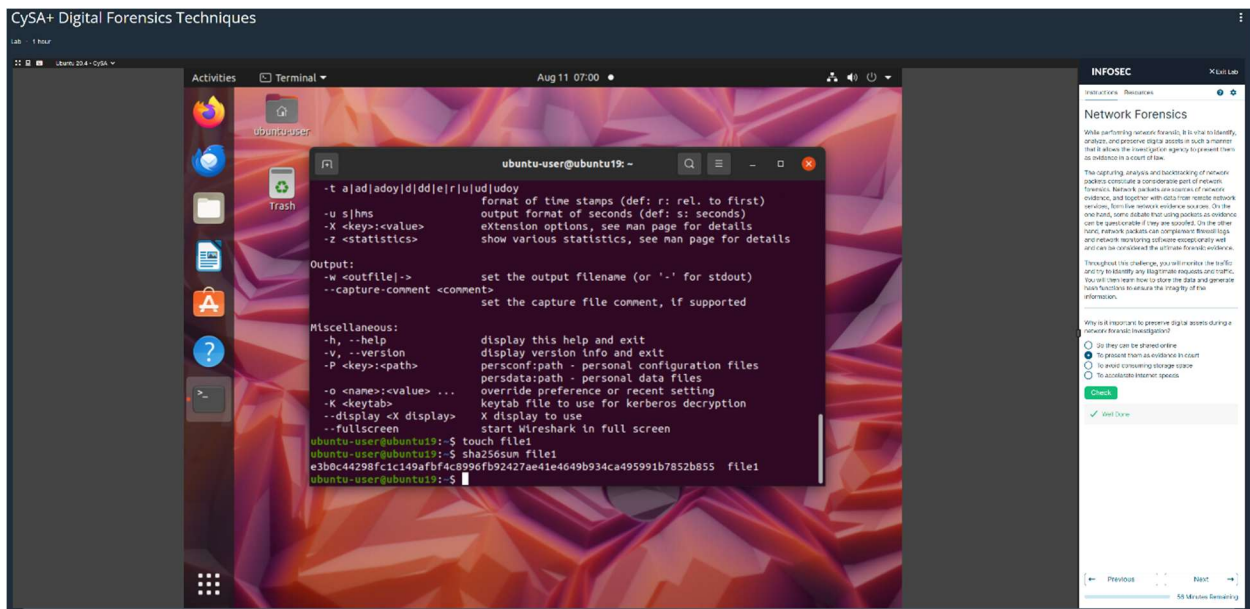


Figure 3: Network Forensics Explanation

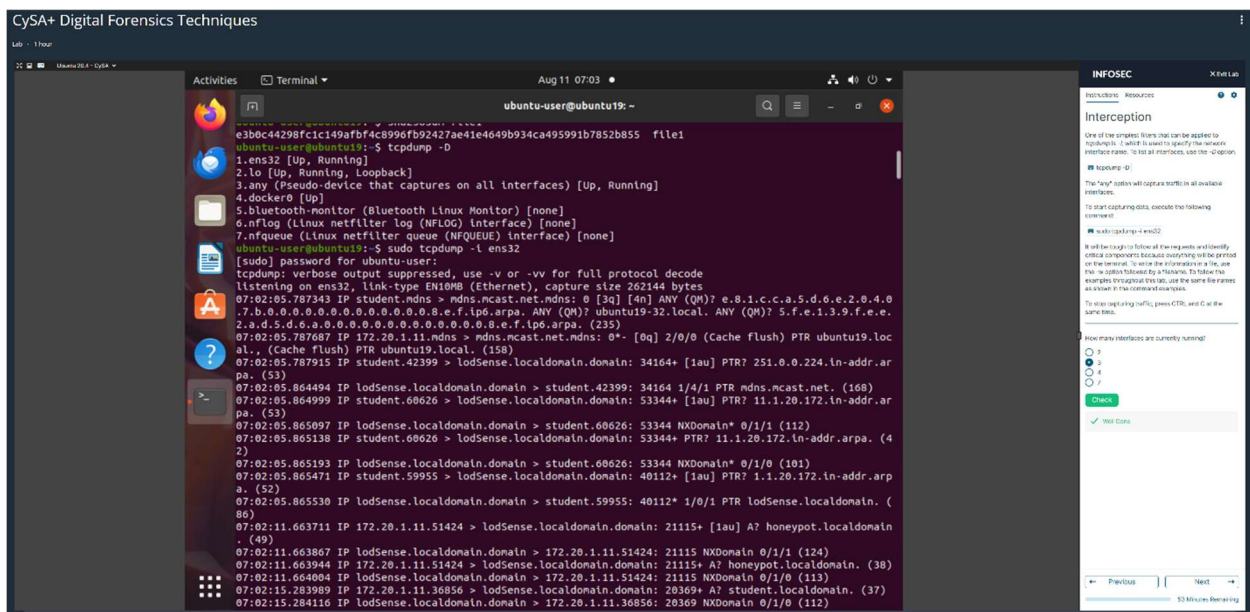


Figure 4: Packet Interception

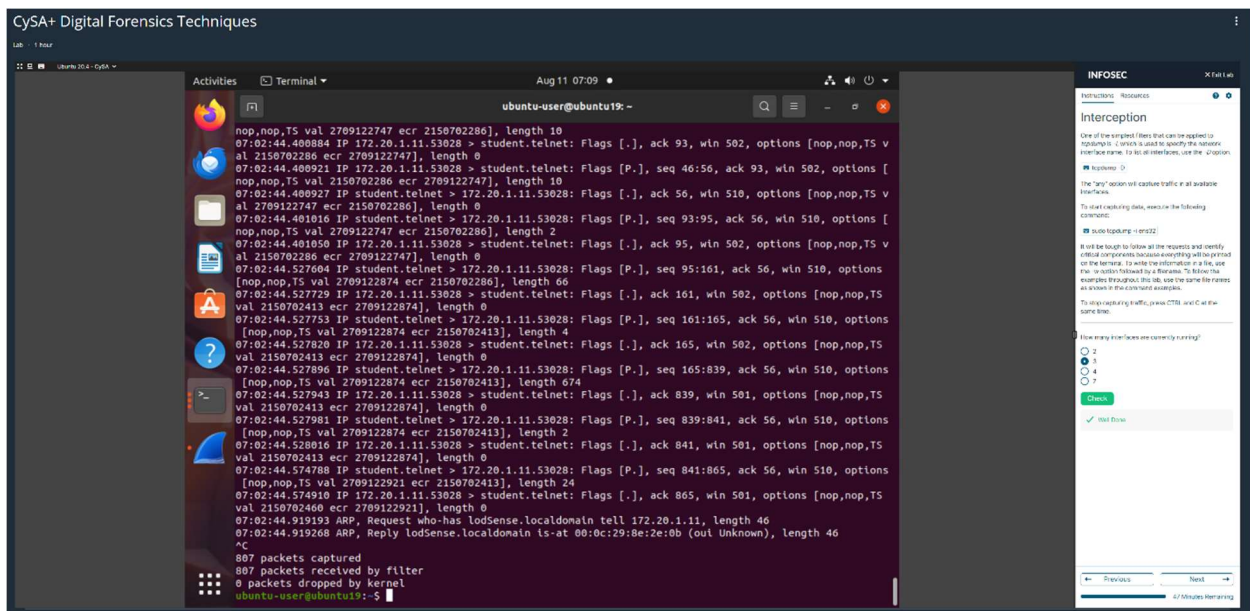


Figure 5: Interception Complete

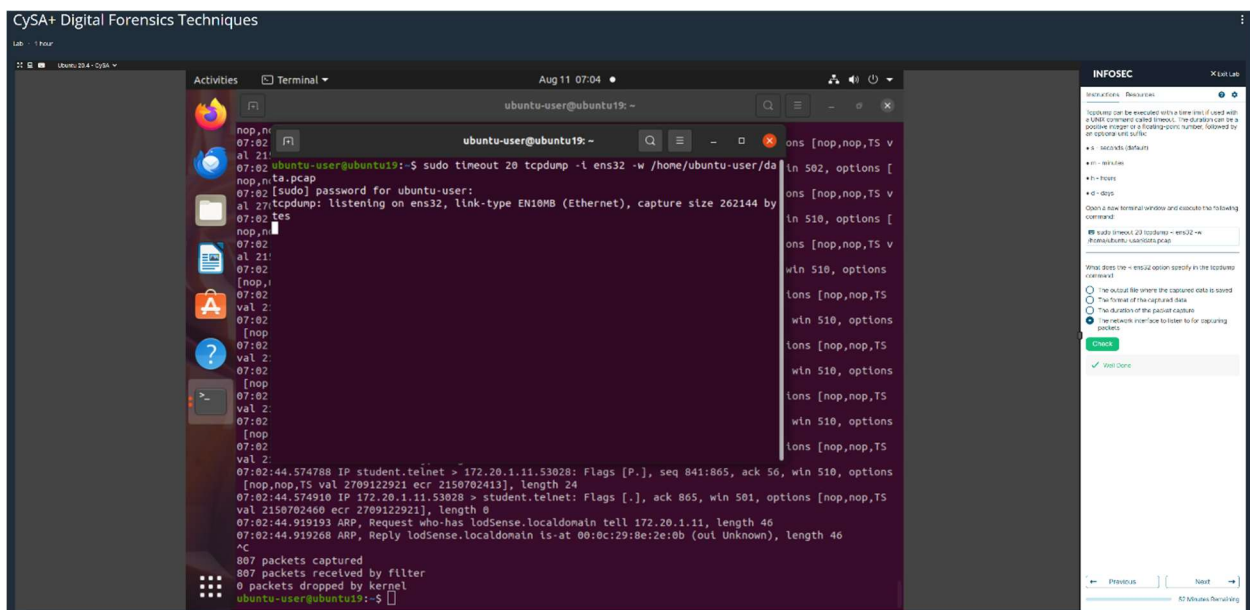


Figure 6: Bounded TCPDump

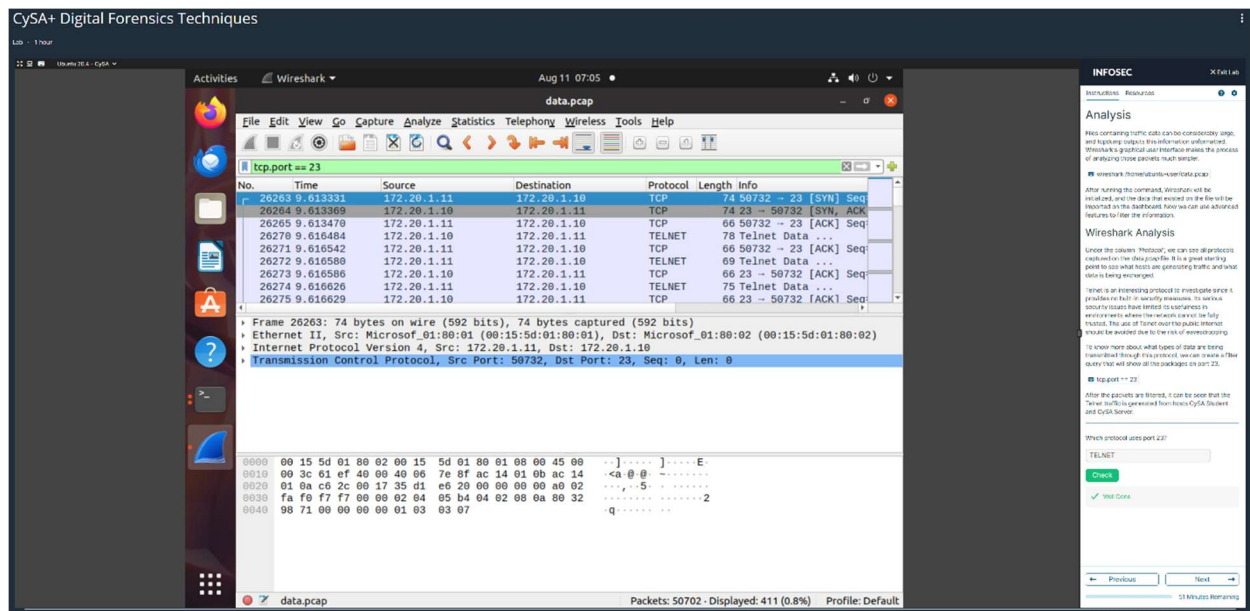


Figure 7: Packet Analysis

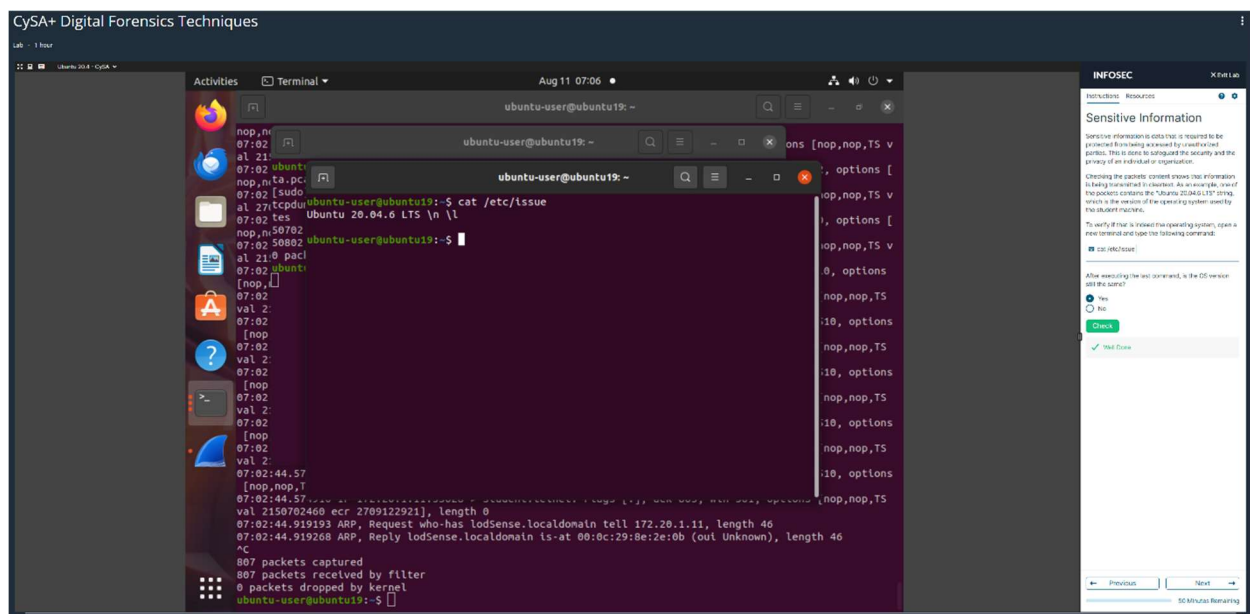


Figure 8: Operating System Identification

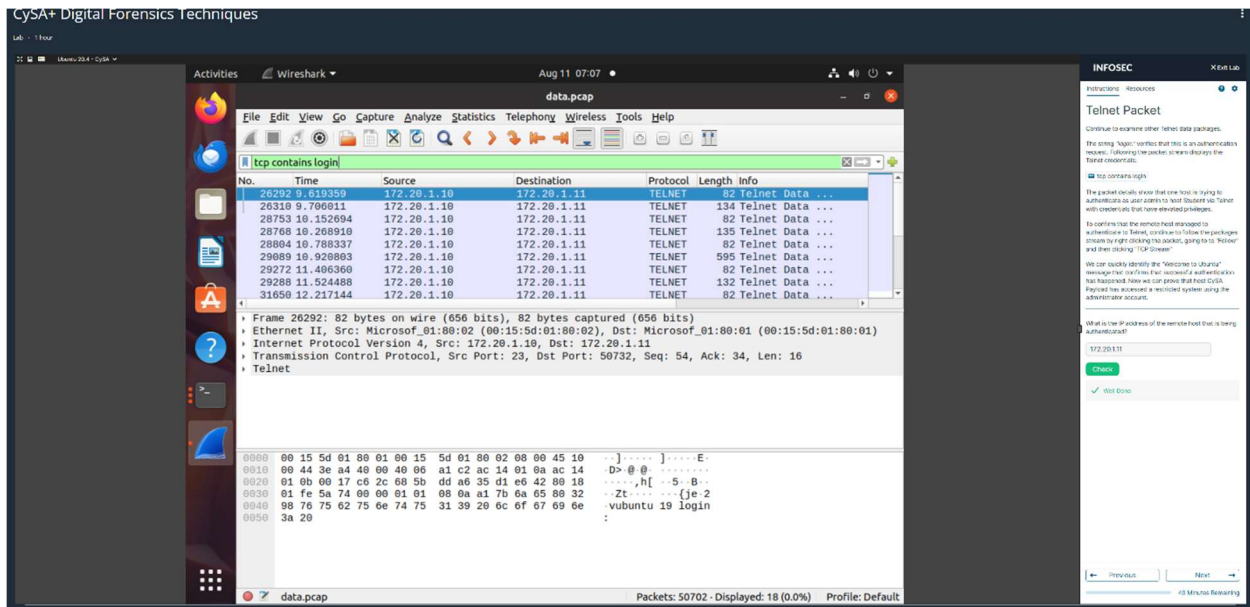


Figure 9: Telnet Packet Analysis

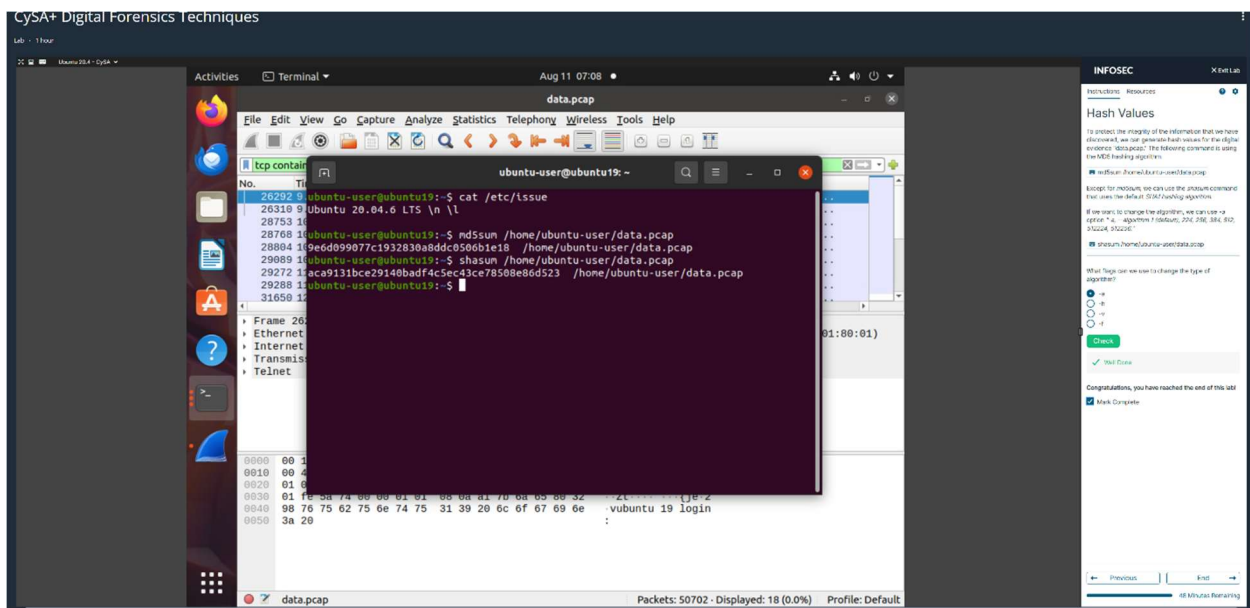


Figure 10: Hash Value Review

References

Callaghan, P. (n.d.). *Why hash values are crucial in digital evidence authentication*.

<https://blog.pagefreezer.com/importance-hash-values-evidence-collection-digital-forensics>

Gonzalez, S. (2024, October 4). *What are the 5 stages of a digital forensics investigation?* ERMPProtect

Cybersecurity - Cybersecurity | Digital Forensics | Penetration Testing.

<https://ermprotect.com/blog/what-are-the-5-stages-of-a-digital-forensics-investigation/>

SolarWinds. (n.d.). *What is packet capture (PCAP)?* SolarWinds IT Glossary. Retrieved August 11, 2025,

from <http://solarwinds.com/resources/it-glossary/pcap>

University of Pittsburgh. (2025, July 2). *Security concerns when using Telnet and FTP*. Information

Technology Knowledge Portal. Retrieved August 11, 2025, from

<https://services.pitt.edu/TDClient/33/Portal/KB/ArticleDet?ID=64>